

CUSTOMER CHURN PREDICTION USING ANN:

PROJECT SUMMARY

1. Project Overview

This project focuses on predicting whether a bank customer will leave (churn) or stay. Using an Artificial Neural Network (ANN), the system analyzes customer data and provides predictions to help businesses retain customers.

2. Business Context

Problem: Customer churn leads to revenue loss and increased customer acquisition costs.

Opportunity: Predicting churn allows businesses to take proactive actions to retain customers.

3. Solution Goals

- Predict customer churn accurately.
- Improve customer retention strategies.
- Support data-driven decision making.

4. Dataset Overview

Key Features:

- Credit Score
- Geography
- Gender
- Age
- Balance
- Estimated Salary

Target Variable:

- Exited (1 = Churn, 0 = No Churn)

5. Data Preprocessing

- Removed irrelevant columns (RowNumber, CustomerId, Surname)
- Encoded categorical data (Gender, Geography)

- Applied feature scaling using StandardScaler

6. Model Building (ANN)

- Sequential neural network model
- Multiple hidden layers with ReLU activation
- Output layer with Sigmoid activation (for binary classification)

7. Model Training

- Optimizer: Adam
- Loss: Binary Crossentropy
- Batch size and epochs used for training

8. Model Evaluation

- Accuracy Score
- Confusion Matrix

9. Deployment

A Streamlit web application was developed where users can input customer details and receive real-time churn predictions.

10. Expected Outcomes

- Improved customer retention
- Faster decision-making
- Reduced operational cost

11. Conclusion

This project demonstrates how Artificial Neural Networks can be effectively used to solve real-world business problems like customer churn prediction. It highlights the importance of data pre-processing, model building, and deployment in delivering actionable insights.